

ARYAN S. SHAH

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EDUCATION

Year	Degree	Institute	CGPA/Marks
2028	B.Tech (Hons) in Ocean Engineering and Naval Architecture	Indian Institute of Technology Kharagpur	7.66
2028	Minor in Artificial Intelligence	Indian Institute of Technology Kharagpur	-
2024	AISSCE	ST. Xavier's Silvassa	94%
2022	AISSE	ST. Xavier's Silvassa	94.2%

ACHIEVEMENTS

- Achieved an all-time highest CodeForces rating of **1754 (Expert)** under the handle [aryan_shah26](https://codeforces.com/profile/aryan_shah26)
- Secured **All India Rank 8170** among 250,000 candidates in JEE Advanced 2024
- Achieved **99.14%ile** in JEE Mains 2024, ranking among the **top 1%** of 1.4 million candidates

PROJECTS

- Automated EDA & AutoML Platform** | *Python, LangGraph, DuckDB, Scikit-learn, Streamlit* [GitHub](#)
- Engineered an out-of-core **EDA engine** using **DuckDB** single-pass **SQL**, reservoir sampling and bounded data-context-object, enabling analysis of large datasets without exhausting memory or LLM context window
 - Built a **LangGraph** tool-using data analysis agent with 20+ registered processes, cost-aware tool routing, concurrent execution and **SQLite backed** conversational state
 - Secured LLM-generated analysis through **AST validation**, **subprocess isolation** and **kernel resource limits**, with optional **Human-In-The-Loop** approval and automatic LLM-driven error correction
 - Developed a deterministic target-health, feature-engineering and sparsity-aware model selection modules, generating model shortlists and Optuna search-space templates from data statistics
 - Benchmarked across 386 automated tests**, achieving 100% tool-level task completion (289/289), 78.1% finding-detection F1, and 100% sandbox attack blocking across 9 attack classes; validated processing on datasets up to ~4.8 GB.
- Large Scale Retail Demand Forecasting** | *Python, Keras/TensorFlow, XGBoost* [GitHub](#)
- Engineered a memory-efficient forecasting pipeline for ~**60M** retail demand records, using dtype downcasting, categorical encoding and partitioned parquet to enable **out-of-core** dataset construction and selective loading
 - Built a GPU accelerated **XGBoost** forecasting model with **QuantileDMatrix**, categorical features and multi-scale lag/rolling statistics, achieving **0.697 WRMSSE** on a 28-day M5 validation horizon
 - Developed a **dual-input LSTM** combining 28-day temporal sequences with learned categorical embeddings implemented per-series train-only normalization and custom sequence generator processing **40M+** training windows
- Production RAG Chat Engine** | *Python, LangGraph, FastAPI, ChromaDB, BM25, Transformers* [GitHub](#)
- Engineered a **deployable RAG system** supporting PDF/HTML/TXT/Markdown ingestion, persistent multi-document collections, table extraction, metadata-aware chunking and citation-grounded generation behind **FastAPI** service
 - Designed a hybrid-retrieval pipeline combining **MiniLM** dense embeddings and **BM25** sparse search via Reciprocal Rank Fusion with cross-encoder reranking, using LangGraph retriever-critic loop that retries low-confidence retrievals
 - Designed a **citation-verification** and **hallucination-filtering** pipeline, to validate claims against retrieved source
 - Developed retrieval and generation evaluation with **Recall@K**, **MRR**, **Context Precision** and **LLM-as-a-judge** faithfulness, alongside response latency, citation-accuracy and reranker score breakdown

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Design Head, Gujarat Cultural Association** - Led 50+ members and organized campus-wide events attracting 1,000+ attendees
- Senior Executive, AI & Metaverse Team, KodeinKGP** - Organized knowledge-sharing sessions on AI/ML for the members
- Associate Member, KGPIan Game Theory Society** - Participated in game theory discussions and problem-solving sessions.

SKILLS

- Languages and Core skills:** Python, C++, C, SQL, Competitive Programming, Data Structures and Algorithms
- AI and ML Frameworks:** PyTorch, TensorFlow, Keras, Scikit-Learn, Transformers, Numpy, Pandas, LangChain/LangGraph
- Development Tools:** Git/GitHub, Jupyter Notebook, VSCode, AntigraVity

COURSEWORK

- Programming:** Foundations of Large Language Models, Programming and Data Structures, Essentials of Machine Learnings
- Mathematics:** Probability and Statistics, Linear Algebra Complex and Numerical Analysis, Applied Computational Methods